University of Southern Denmark

Pulmonary Valve Replacement in Tetralogy of Fallot

Procedural Volume and Durability of Bioprosthetic Pulmonary Valves

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ORIGINAL RESEARCH

STRUCTURAL

Pulmonary Valve Replacement in Tetralogy of Fallot

Procedural Volume and Durability of Bioprosthetic Pulmonary Valves

Mathis Gröning, MD, PhD,[a,b] Morten Holdgaard Smerup, MD, PhD,[c] Kim Munk, MD, PhD,[d] Helle Andersen, MD, PhD,[e] Dorte Guldbrand Nielsen, MD, PhD,[d] Henrik Nissen, MD, PhD,[f] Ulrik Markus Mortensen, MD, PhD,[d] Annette Schophuus Jensen, MD, PhD,[a] Pernille Steen Bække, MD,[a] Jesper Bjerre, MD,[g] Morten Engholm, MD, PhD,[d] Niels Vejlstrup, MD, PhD,[a] Klaus Juul, MD, PhD,[h] Eva Vad Søndergaard, MD,[f] Hans Gustav Hørsted Thyregod, MD, PhD,[c] Henrik Ørbæk Andersen, MD, PhD, DMSc,[c] Morten Helvind, MD,[c] Ole De Backer, MD, PhD,[a] Christian Jøns, MD, PhD,[a] Michael Rahbek Schmidt, MD, PhD,[a] Troels Højsgaard Jørgensen, MD, PhD,[a] Lars Sondergaard, MD, DMSc[a]

ABSTRACT

BACKGROUND Robust data on changes in pulmonary valve replacement (PVR) procedural volume and predictors of bioprosthetic pulmonary valve (BPV) durability in patients with tetralogy of Fallot (TOF) are scarce.

OBJECTIVES This study sought to assess temporal trends in PVR procedural volume and BPV durability in a nationwide, retrospective TOF cohort.

METHODS Data were obtained from patient records. Robust linear regression was used to assess temporal trends in PVR procedural volume. Piecewise exponential additive mixed models were used to estimate BPV durability, defined as the time from implantation to redo PVR with death as a competing risk, and to assess risk factors for reduced durability.

RESULTS In total, 546 PVR were performed in 384 patients from 1976 to 2021. The annual number of PVR increased from 0.4 to 6.0 per million population ($P < 0.001$). In the last decade, the transcatheter PVR volume increased by 20% annually ($P < 0.001$), whereas the surgical PVR volume did not change significantly. The median BPV durability was 17 years (Q1: 10-Q3: 10 years-not applicable). There was no significant difference in the durability of different BPV after adjustment for confounders. Age at PVR (HR: 0.78 per 10 years from <1 year; 95% CI: 0.63-0.96; $P = 0.02$) and true inner valve diameter (9-17 mm vs 18-22 mm HR: 0.40; 95% CI: 0.22-0.73; $P = 0.003$ and 18-22 mm vs 23-30 mm HR: 0.59; 95% CI: 0.25-1.39; $P = 0.23$) were associated with reduced BPV durability in multivariate models.

CONCLUSIONS The PVR procedural volume has increased over time, with a greater increment in transcatheter than surgical PVR during the last decade. Younger patient age at PVR and a smaller true inner valve diameter predicted reduced BPV durability. (J Am Coll Cardiol Intv 2024;17:217–227) © 2024 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

From the [a]Department of Cardiology, Rigshospitalet, Copenhagen University Hospital, Copenhagen, Denmark; [b]Department of Cardiology, Zealand University Hospital, Roskilde, Denmark; [c]Department of Cardio-Thoracic Surgery, Rigshospitalet, Copenhagen University Hospital, Copenhagen, Denmark; [d]Department of Cardiology, Aarhus University Hospital, Aarhus, Denmark; [e]Hans Christian Andersen Children's Hospital, Odense University Hospital, Odense, Denmark; [f]Department of Cardiology, Odense University Hospital, Odense, Denmark; [g]Department of Pediatrics, Aarhus University Hospital, Aarhus, Denmark; and the [h]Department of Pediatrics, Rigshospitalet, Copenhagen University Hospital, Copenhagen, Denmark.

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ABBREVIATIONS AND ACRONYMS

BPV = bioprosthetic pulmonary valve

PVR = pulmonary valve replacement

RV = right ventricle

RVOT = right ventricular outflow tract

TOF = tetralogy of Fallot

TPV = transcatheter pulmonary valve

Pulmonary valve replacement (PVR) is one of the most common reinterventions performed in patients with congenital heart disease after initial surgical repair, and the number of procedures is expected to rise.[1] The majority of patients undergoing PVR are born with tetralogy of Fallot (TOF).[2] One of the most important knowledge gaps in the current management of patients with TOF is the optimal timing of PVR in patients who develop severe pulmonary regurgitation after intracardiac repair.[3] In patients with severely dilated right ventricle (RV) or reduced RV function, PVR has been associated with reduced mortality and a lower incidence of ventricular arrhythmia.[4] However, the prognostic benefit of PVR in patients without severe RV dilatation and preserved RV function remains a subject of debate.[4] Although early PVR may be beneficial in terms of RV preservation, important concerns are the potential increase in reinterventions required throughout a lifetime and earlier exposure to the risk of prosthetic valve endocarditis.[3] Furthermore, younger age at the time of PVR is one of the most commonly reported risk factors for reduced bioprosthetic pulmonary valve (BPV) durability.[5-12] Another important controversy is the optimal BPV choice.[13] So far, only a few studies have found significant variation in the durability of different BPV.[6,7,9,14,15] An important weakness of previous studies evaluating BPV durability is that the underlying cohorts consisted of patients with different congenital heart diseases. The aim of this study was to assess temporal trends in PVR procedural volume and BPV durability based on a nationwide TOF cohort.

METHODS

The Danish National Patient Registry and the EVITRA (Early Versus Later Re-valving in Tetralogy of Fallot with Free Pulmonary Regurgitation; NCT04084132) study were used to identify patients with TOF who underwent PVR.[16] Patient records and surgical reports were reviewed up until December 3, 2021, to obtain data including BPV type and inner valve diameters. Patients with double outlet right ventricle of Fallot type were included in the cohort. Patients with an atrioventricular septal defect, absent pulmonary valve syndrome, and pulmonary atresia with a ventricular septal defect and major aortopulmonary collateral arteries were excluded. The main outcome was BPV durability, which was defined as the time from implantation to surgical or transcatheter redo PVR. The first PVR was defined as surgical or transcatheter PVR in a patient with a native or patched right ventricular outflow tract (RVOT) either during or after intracardiac repair. Redo PVR was defined as any subsequent surgical or transcatheter PVR. True inner diameters were used for Perimount/Magna/Magna Ease valves (Edwards Lifesciences) as per the information provided by the manufacturer, Hegar dilator measurements were used for homografts and Contegra grafts (Medtronic), and balloon sizes were used for transcatheter pulmonary valve (TPV). True inner diameters or diameters provided in the surgical reports were used for other BPV depending on the available information. The study was approved by the Danish Patient Safety Authority (record number: R-20061426) and the Danish Data Protection Agency (record number: P-2020-1129).

STATISTICAL ANALYSIS. Continuous data are presented as median (Q1-Q3), categoric data as counts with percentages, and cumulative incidence probabilities as percentages with 95% CIs. Continuous and categoric data were compared across groups using the Kruskal-Wallis and chi-square tests, respectively. The overall crude median BPV durability was estimated using the Aalen-Johansen estimator. Piecewise exponential additive mixed models were used to estimate cumulative incidence probabilities for the replacement of different BPV options and according to patient age at PVR as well as the true inner valve diameter.[17] Piecewise exponential additive mixed models and cause-specific Cox regression were used to identify risk factors for reduced BPV durability in multivariate models. Variable selection for the multivariate models was based on previously published data and relevance of the individual variables from a clinical perspective. Death was considered as a competing risk whenever the Aalen-Johansen estimator or piecewise exponential additive mixed models were used. A frailty term was included in the piecewise exponential additive mixed models and Cox regression models to account

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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TABLE 1 Baseline Characteristics

Patient Characteristics	Overall (N = 384)	Transannular Patch (n = 212)	RV-PA Conduit (n = 94)	Valve-Sparing Repair (n = 78)	*P* Value
Male	239 (62)	134 (63)	46 (49)	59 (76)	0.05
Palliative shunt	133 (35)	37 (17)	71 (76)	25 (32)	<0.001
Age at shunt palliation, mo	5 (0-30)	5 (1-32)	1 (0-18)	23 (9-40)	<0.001
Intracardiac repair	384 (100)	212 (100)	94 (100)	78 (100)	1.00
Age at intracardiac repair, y	2 (1-6)	2 (1-5)	1 (1-5)	5 (2-9)	<0.001
Time from intracardiac repair to first PVR	17 (11-30)	17 (12-26)	7 (1-12)	33 (22-38)	<0.001

Bioprosthetic Pulmonary Valve Characteristics	Overall (N = 546)	Homograft (n = 268)	Contegra (n = 90)	Transcatheter (n = 83)	Perimount/Magna/ Magna Ease (n = 70)	Other (n = 35)	*P* Value
Patient age at implantation, y	22 (17) 18 (10-35)	24 (17) 19 (12-37)	8 (9) 6 (1-11)	32 (15) 30 (18-43)	28 (13) 24 (19-33)	18 (19) 12 (3-23)	<0.001
True inner valve diameter	22 (4) 22 (20-24)	22 (3) 22 (21-25)	17 (3) 18 (14-20)	23 (2) 23 (23-25)	22 (2) 22 (22-22)	21 (5) 22 (21-22)	<0.001

Values are n (%) or median (Q1-Q3). *P* values were derived from omnibus tests (Kruskal-Wallis tests for continuous data and chi-square tests for categoric data) to assess differences across the different groups. Post hoc between-group comparisons are provided in Supplemental Table 2.

PVR = pulmonary valve replacement; RV-PA = right ventricle-to-pulmonary artery.

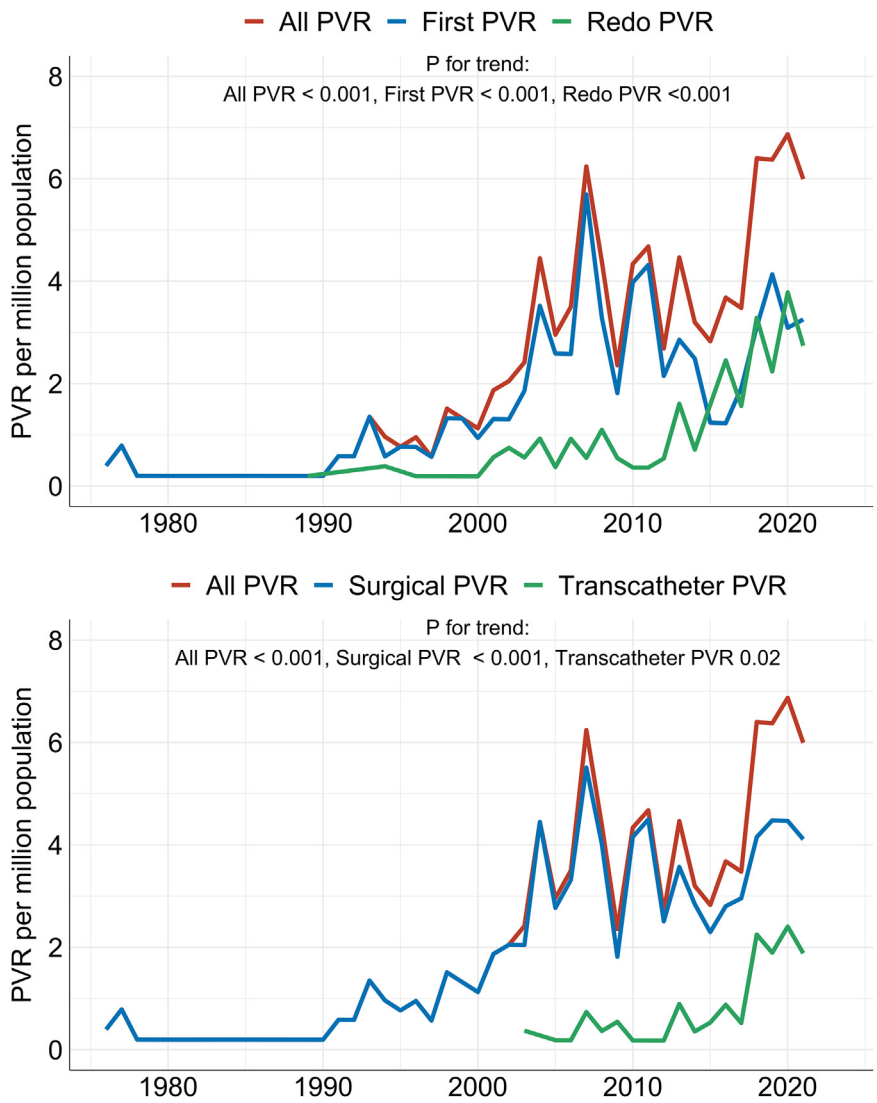
for multiple PVR performed in the same patient. The proportional hazards assumption was evaluated using complementary log-log curves, cumulative martingale residual plots, the cumulative score process test for proportionality, and assessment of independence between scaled Schoenfeld residuals and time for each covariate. No violations of the proportional hazards assumption were detected. The reverse Kaplan-Meier method was used to calculate the median follow-up after PVR. The Mann-Kendall test was used to assess the significance of temporal trends in PVR procedural volume over the entire study period. Robust linear regression with a modified M-type estimator was used to obtain estimates for the annual increase in PVR procedural volume.[18] Multiple imputation of missing data (Supplemental Table 1) was performed with an algorithm based on chained random forests.[19] Predictive mean matching was used to improve the quality of imputed values. A total of 100 complete data sets were created. The data sets were pooled using the median of the 100 imputed values across the complete data sets. The quality of the imputed values was assessed graphically. *P* values were considered statistically significant at the 2-sided <0.05 level. R Statistics (R Foundation for Statistical Computing) was used to perform the statistical analyses. Add-on packages used to perform the analyses outlined in this paper are provided in the Supplemental Appendix.

RESULTS

A total of 546 PVR, 463 surgical and 83 transcatheter PVR, were performed in 384 patients with TOF in Denmark between 1976 and 2021. Baseline characteristics are displayed in **Table 1** and Supplemental Table 2. The median follow-up after PVR was 10.4 years (Q1-Q3: 3.6-16.5 years). A total of 162 BPV (30%) were replaced, and 41 patients (11%) who had undergone PVR died during follow-up. Indications for the replacement of BPV are provided in Supplemental Table 3 and Supplemental Figure 1.

TEMPORAL TRENDS IN PROCEDURAL VOLUME. Between 1976 and 2021, the annual number of PVR increased from 0.4 to 6.0 per million population ($P < 0.001$) (**Figure 1**). The estimated annual increase in PVR during this period was 13% (95% CI: 7%-18%; $P < 0.001$) with a lower increase in procedural volume during the beginning and a higher increase toward the end of the study period (**Figure 1**, Supplemental Table 4). First and redo PVR as well as surgical and transcatheter PVR increased significantly over time (**Figure 1, Central Illustration**, Supplemental Table 4). During the last decade, the annual number of transcatheter PVR increased by 20% (95% CI: 15%-26%; $P < 0.001$), whereas there was no significant increase in the annual number of surgical PVR (annual increase of 6%; 95% CI: −8% to 21%; $P = 0.41$) (Supplemental Table 4). The number of patients alive with a BPV in

FIGURE 1 Temporal Trends in the Number of PVR Performed Among Patients With Tetralogy of Fallot in Denmark

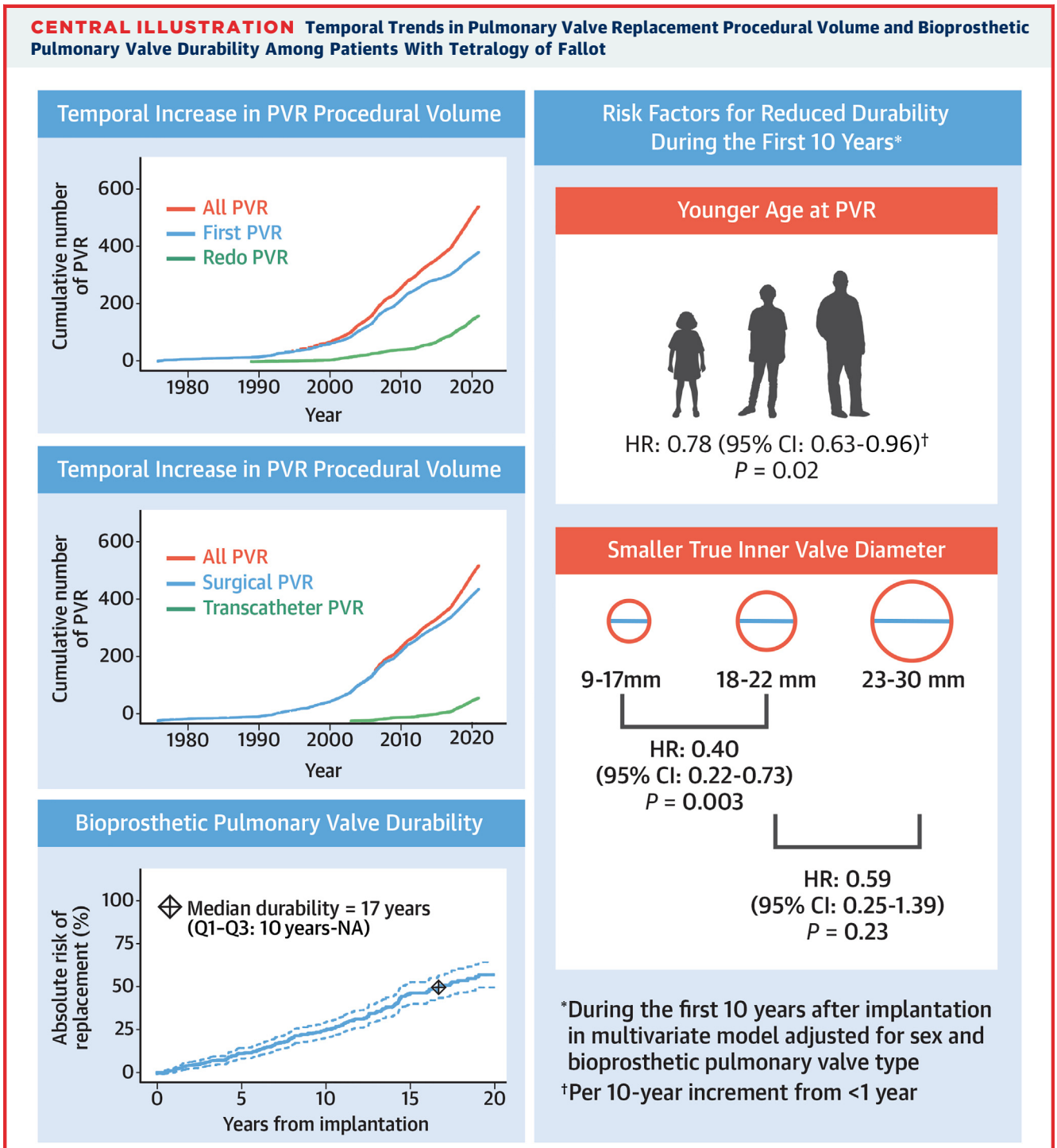


The number of annual pulmonary valve replacement (PVR) per million population over time divided into (upper panel) first and redo PVR and (lower panel) surgical and transcatheter PVR (lower panel). The first PVR was defined as surgical or transcatheter PVR in a patient with a native or patched right ventricular outflow tract either during or after intracardiac repair. Redo PVR was defined as any subsequent surgical or transcatheter PVR.

situ increased from <1 to 59 per million population between 1976 and 2021 (**Figure 2**).

BPV DURABILITY. The overall crude median BPV durability was 17 years (Q1-Q3: 10 years-not applicable) (**Central Illustration**). Of the implanted BPV, 268 (49%) were homografts (pulmonary: n = 120, aortic: n = 101, unknown: n = 47), 90 (17%) were Contegra grafts, 70 (13%) were Perimount/Magna and Magna Ease valves, 83 (15%) were TPV (Melody [Medtronic]: n = 64, SAPIEN [Edwards]: n = 19), and 35 (6%) were other BPV. The most common BPV defined as other were Hancock valves (Medtronic; n = 7, 20%) and Freestyle valves (Medtronic; n = 5, 14%). Supplemental Figure 2 illustrates the use of these BPV in Denmark over time. The cumulative incidence of replacement 10 years after implantation was 14% for Perimount/Magna and Magna Ease valves, 20% for homografts, 29% for TPV, and 47% for Contegra grafts (**Figure 3**). TPV were more commonly replaced

CENTRAL ILLUSTRATION Temporal Trends in Pulmonary Valve Replacement Procedural Volume and Bioprosthetic Pulmonary Valve Durability Among Patients With Tetralogy of Fallot



Gröning M, et al. J Am Coll Cardiol Intv. 2024;17(2):217–227.

(Left) The cumulative number of pulmonary valve replacement (PVR) per million population over time divided into first and redo PVR as well as surgical and transcatheter PVR. (Right upper panel) The overall crude bioprosthetic pulmonary valve durability estimated as the cumulative incidence of replacement according to years from implantation using the Aalen-Johansen estimator with death as a competing risk. (Right lower panel) Risk factors for reduced bioprosthetic pulmonary valve durability within the first 10 years after implantation in a multivariate piecewise exponential additive mixed model including sex, age at PVR, true inner valve diameter, and bioprosthetic pulmonary valve type with death as a competing risk. NA = not applicable.

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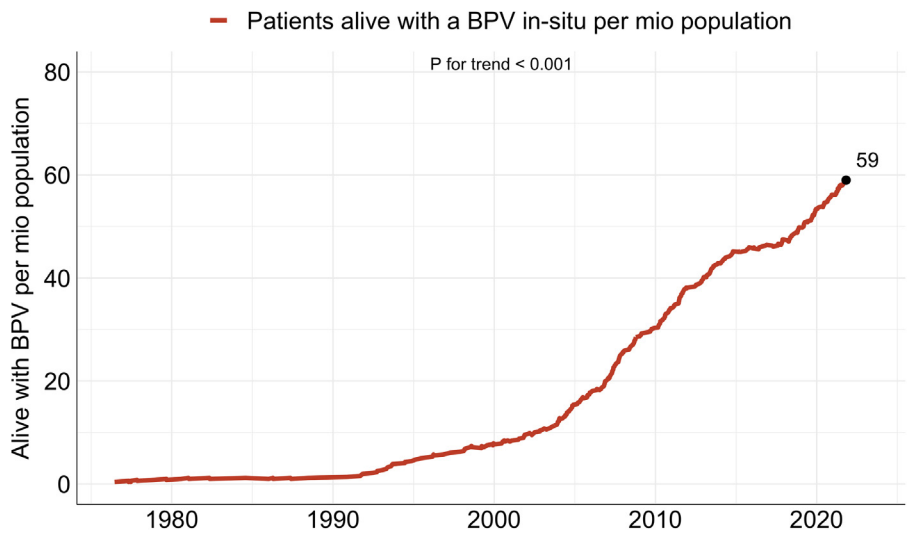
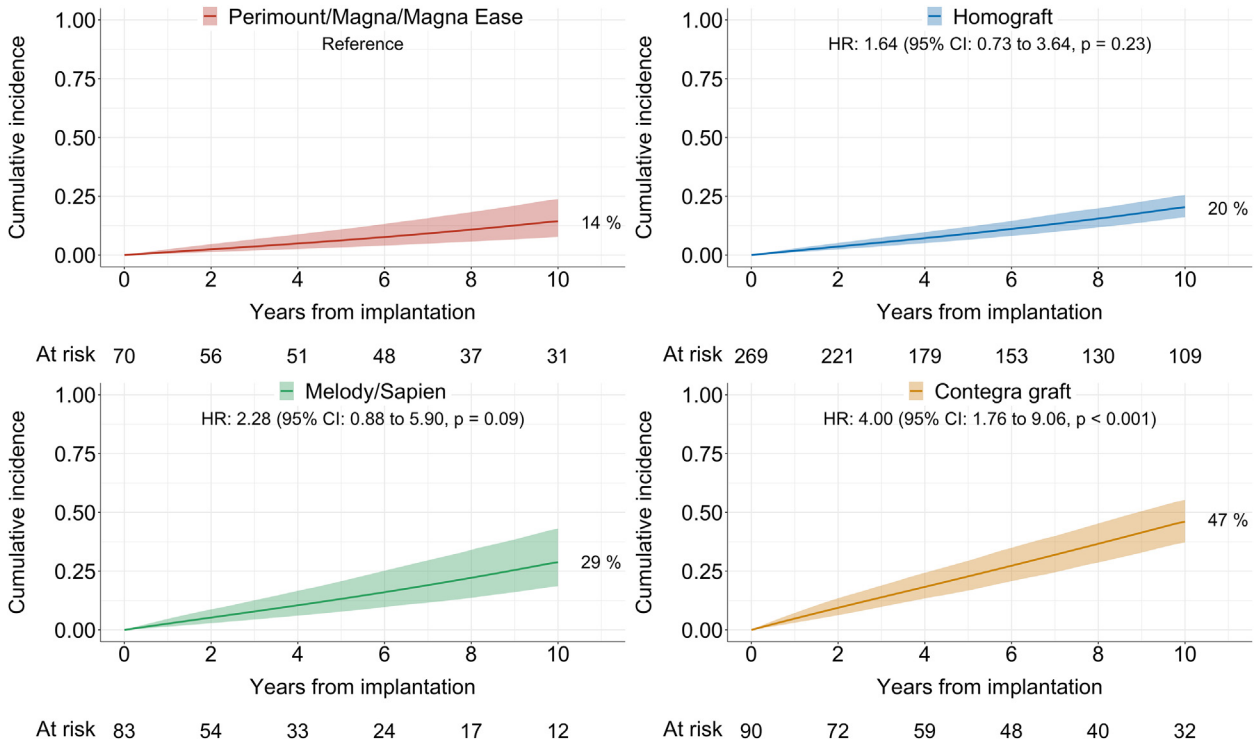
FIGURE 2 Temporal Trend in the Number of Patients Living With a Bioprosthetic Pulmonary Valve In Situ



The number of patients living with a bioprosthetic pulmonary valve (BPV) in situ per million inhabitants in Denmark cumulated over time. The cumulative count was updated at each date with an event, either implantation of a bioprosthetic pulmonary valve or death of a patient who underwent previous pulmonary valve replacement. mio = million.

FIGURE 3 Crude Durability of Different Bioprosthetic Pulmonary Valves



Estimated cumulative incidence function curves (solid line) with pointwise 95% confidence limits (color shading) illustrating the time-dependent incidence of replacement for different bioprosthetic pulmonary valves without adjustment for patient age at implantation and true inner valve diameter. Death was accounted for as a competing risk.

TABLE 2 Predictors of Bioprosthetic Pulmonary Valve Durability Within the First 10 Years After Implantation Estimated Using Piecewise Additive Exponential Models

	Univariate		Multivariate	
	HR (95% CI)	*P* Value	HR (95% CI)	*P* Value
Male	1.54 (1.00-2.37)	0.05	1.49 (0.96-2.30)	0.07
Shunt palliation	1.01 (0.68-1.51)	0.96		
Age at PVR[a]	0.86 (0.75-0.96)	0.02	0.78 (0.63-0.96)	0.02
Age at PVR, y				
0-4	Reference	Reference		
5-20	0.60 (0.38-0.95)	0.03		
>20	0.25 (0.14-0.44)	<0.001		
True inner valve diameter[b]	0.91 (0.87-0.95)	<0.001		
True inner valve diameter, mm				
9-17	Reference	Reference	Reference	Reference
18-22	0.26 (0.16-0.42)	<0.001	0.40 (0.22-0.73)	0.003
23-30	0.21 (0.12-0.37)	<0.001	0.59 (0.25-1.39)	0.23
BPV type				
Perimount/Magna	Reference	Reference	Reference	Reference
Homograft	1.63 (0.73-3.64)	0.23	1.38 (0.60-3.18)	0.44
Transcatheter	2.28 (0.88-5.88)	0.09	2.77 (0.99-7.69)	0.05
Contegra	3.99 (1.76-9.04)	<0.001	1.98 (0.76-5.16)	0.17
Other	2.92 (1.02-8.32)	0.05	2.67 (0.89-8.00)	0.08

[a]10-year increment from <1 year. [b]1-mm increment from 11 mm.
BPV = bioprosthetic pulmonary valve; PVR = pulmonary valve replacement.

because of infective endocarditis than other BPV (Supplemental Table 3, Supplemental Figure 1). There was no significant difference in the durability of aortic and pulmonary homografts (Supplemental Figure 3). The overall median patient age at PVR was 18 years (Q1: 10-Q3: 35 years). There was a significant association between younger patient age at PVR as well as a smaller inner valve diameter and reduced BPV durability (**Table 2**, Supplemental Table 5). The cumulative incidence of replacement after 10 years for BPV implanted in patients who were 0 to 4 years of age, 5 to 20 years of age, or older than 20 years of age at the time of PVR was 45%, 30%, and 15%, respectively (**Figure 4**). The overall median true inner diameter of the implanted BPV was 22 mm (Q1-Q3: 20-24 mm). The cumulative incidence of replacement after 10 years for BPV with a true inner diameter of 9 to 17 mm, 18 to 22 mm, or 23 to 30 mm was 54%, 23%, and 19%, respectively (**Figure 4**). **Figure 5** illustrates the true inner diameter of the implanted BPV according to patient age at PVR. Differences in the cumulative incidence of replacement for the different BPV types did not remain statistically significant after adjustment for true inner valve diameters and year of implantation (Supplemental Figure 3). Younger age at PVR and a smaller true inner valve diameter were associated with significantly reduced BPV durability in multivariate models adjusted for sex and BPV type (**Central Illustration**, **Table 2**, Supplemental Table 5).

DISCUSSION

To our knowledge, this is the largest assessment of trends in PVR procedural volume and BPV durability based on a nationwide TOF cohort. The main findings were as follows: 1) the PVR procedural volume increased considerably over the last 45 years; 2) the median BPV durability was 17 years (Q1-Q3: 10 years-not applicable); 3) younger age at PVR and a smaller true inner valve diameter were associated with reduced BPV durability; and 4) there was no significant variation in the durability of different BPV after adjustment for confounders.

PROCEDURAL VOLUME. To our knowledge, only 2 other studies have addressed temporal changes in PVR procedural volume. O'Byrne et al[20] reported an increase in the number of PVR performed from 2004 to 2012 based on the Pediatric Health Systems Information Database, which captures data from 43 tertiary pediatric hospitals in the United States. In their study, 799 patients underwent PVR with an annual increase of 12 procedures per year. Egbe et al[21] reported an increase in the number of PVR in patients 18 years of age or older from 2000 to 2014

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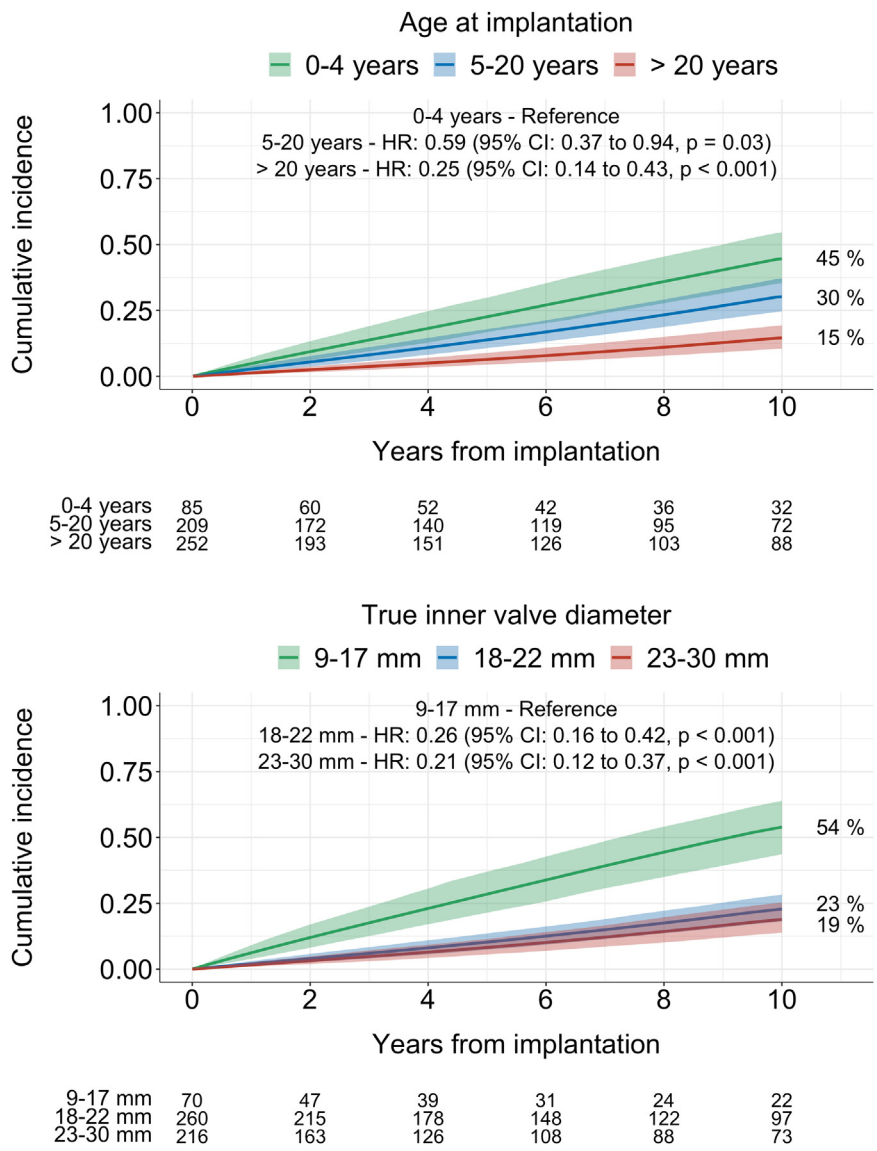
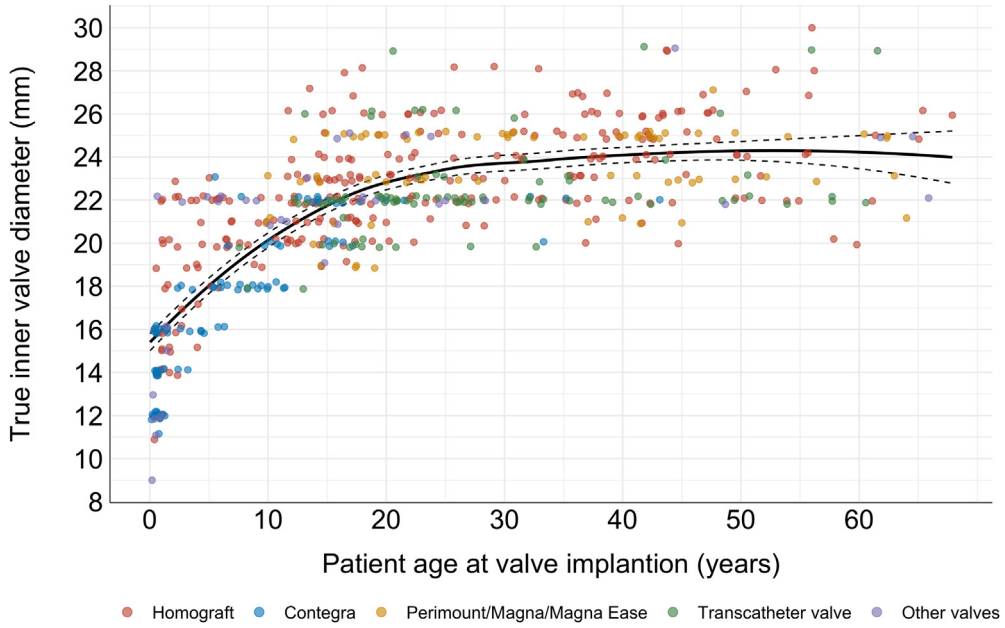
FIGURE 4 Durability of Bioprosthetic Pulmonary Valves According to Patient Age at Implantation and True Inner Valve Diameter



Estimated cumulative incidence function curves (solid line) with pointwise 95% confidence limits (color shading) illustrating the time-dependent incidence of replacement for bioprosthetic pulmonary valves implanted at different ages (upper panel) and with different true inner valve diameters (lower panel). Death was accounted for as a competing risk.

based on the National Inpatient Sample, representing the largest all-payer database of hospital inpatient stays in the United States, which contains data from a 20% sample of nationwide data. In their study, 1,230 of 18,353 admissions (6.7%) of adult patients with TOF were admissions for PVR. The proportion of admission for PVR relative to the total number of admissions was 2.7% from 2000 to 2004, 6.3% from 2005 to 2009, and 9.6% from 2010 to 2014. Neither Byrne et al[20] nor Egbe et al[21] provided procedural volumes indexed to population size, which prohibits direct comparison with our data. Improved survival of patients with severe RVOT obstruction who undergo intracardiac repair with a transannular patch, the implementation of guidelines for PVR in asymptomatic patients with severe pulmonary regurgitation and better evidence for the prognostic benefit of the procedure, more use of cardiac magnetic resonance allowing for better assessment of RV volumes and function than

FIGURE 5 True Inner Diameter of the Implanted Bioprosthetic Pulmonary Valves According to Patient Age at Implantation



A scatterplot depicting the true inner diameter of implanted bioprosthetic pulmonary valves according to patient age at implantation. Each point represents 1 valve implanted during or after intracardiac repair. A small amount of random vertical and horizontal variation has been added to each point in order to avoid overplotting and aid visual assessment. The solid line represents smoothed conditional means predicted using locally estimated scatterplot smoothing with 95% CI (dashed lines).

echocardiography, reduced mortality associated with PVR, and the introduction of transcatheter pulmonary valve replacement may have contributed to the increasing PVR procedural volume.[3,4,22,23]

BPV DURABILITY. In line with previous studies, we found younger age at PVR and a smaller true inner BPV diameter to be associated with reduced BPV durability.[5-12,14,24] Unfavorable flow dynamics, somatic outgrowth, and stronger immunologic reaction to biological material in younger patients are the proposed mechanisms leading to faster degeneration of BPV in younger patients and BPV with smaller inner diameters.[6,25,26] Whether BPV oversizing leads to improved BPV durability remains controversial, which may explain why we did not see significant improvement in the durability of BPV with a diameter beyond 22 mm in the multivariate models. Jang et al[26] reported reduced longevity of BPV with a z-score >2.0 compared with BPV with a z-score ≤2.0, and Askovich et al[27] reported reduced longevity of homografts with a z-score >2.7 compared with homografts with a z-score <2.7. Furthermore, Chen et al[10] and Baird et al[14] reported an association between a larger inner BPV diameter indexed to patient body surface area and reduced BPV durability. In contrast, Willetts et al[28] reported a progressive increase in BPV durability with increasing z-scores ranging from −1 to 3, and Karamlou et al[25] reported smaller z-scores to be associated with reduced BPV durability, although no reduction in the risk of replacement was achieved if z-scores exceeded 1.0. Interestingly, Karamlou et al reported that z-scores <1.0 were associated with a faster increase in transvalvular gradients, and z-scores >3.0 were associated with a faster increase in transvalvular gradients as well as regurgitation. Hence, BPV oversizing may only be beneficial in specific age groups and with specific BPV.

Variation in the performance of different BPV is of particular interest because, unlike other factors, the BPV choice is modifiable. So far, only a few studies have reported significant differences in the durability of different BPV.[6,7,9,14,15] We did not find any significant difference in the durability of BPV included in the study after adjustment for confounders. However, there was a considerable difference in the numeric difference in replacement rates at 10 years. Thus, although other factors, such as age at PVR and inner valve diameter may be of greater importance for BPV durability, our study could also have been

underpowered to show a difference in the durability of the included BPV. Replacement rates for the BPV in our study were roughly in line with previously published data from comparable studies in which patients with TOF comprised the majority of the study cohort.[7,9,11,26,29-31] Recent studies have demonstrated particularly low replacement rates for decellularized homografts.[31,32] However, the experience with these BPV is still limited. Our data indicated a considerable increase in the number of transcatheter PVR during the last decade. Novel TPV platforms designed specifically for implantation in dilated RVOTs have recently been introduced.[33-36] Assessment of their durability, risk of infective endocarditis, and complications associated with subsequent reinterventions will be important future research topics.

STUDY STRENGTHS AND LIMITATIONS. Important strengths of this study are the nationwide cohort consisting exclusively of patients with TOF and the long follow-up. Because of the retrospective nature of this study, bias caused by unavailable data or unrecognized confounding may persist. This study was concerned with durability measured as the time to replacement. However, BPV dysfunction may occur long before replacement is performed; thus, the provided durability estimates are likely inflated relative to durability measured as the time to BPV dysfunction. We were unable to obtain patient weight, height, and pulmonary valve *z*-scores at the time of PVR; therefore, we were unable to evaluate the impact of BPV oversizing or undersizing on durability. Despite thorough data collection efforts, a certain amount of data remained missing (Supplemental Table 1) and was imputed using multiple imputation techniques. Supplemental Figures 5 to 10 provide an overview of the imputation quality.

CONCLUSIONS

The PVR procedural volume has increased over time, with a greater increment in transcatheter than surgical PVR during the last decade. Younger patient age at PVR and a smaller true inner valve diameter were associated with reduced BPV durability. There was no significant variation in the durability of different BPV after adjustment for confounders.

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ADDRESS FOR CORRESPONDENCE: Dr Mathis Gröning, The Heart Centre, Rigshospitalet Department of Cardiology, Section 2011, Blegdamsvej 9, 2001 Copenhagen, Denmark. E-mail: mathis.groening@regionh.dk.

PERSPECTIVES

WHAT IS KNOWN? The optimal timing of PVR is one of the most important knowledge gaps in the management of patients with TOF. Earlier PVR may lead to better RV preservation. An important concern is the potential increase in PVR required throughout a lifetime; thus, data regarding risk factors for reduced BPV durability are needed.

WHAT IS NEW? The PVR procedural volume increased considerably over the last 45 years. Younger age at PVR and smaller inner valve diameters were associated with reduced BPV durability.

WHAT IS NEXT? During the last decade, the largest increase in the PVR procedural volume was for transcatheter procedures. Data regarding the durability, risk of endocarditis, and risk associated with subsequent reinterventions for recently approved novel TPV will be an important research area.

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APPENDIX For supplemental tables and figures, please see the online version of this paper.